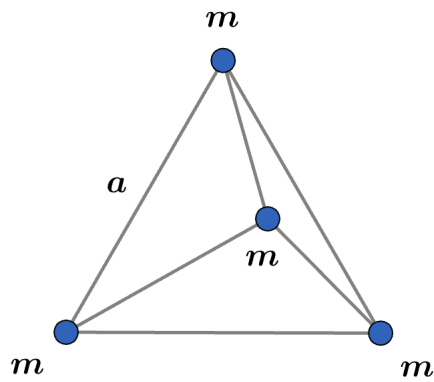


# 2010 F=ma Exam: Problem 17

Kevin S. Huang



Each pair of masses contributes

$$U = -\frac{Gm^2}{a}$$

gravitational potential energy to the arrangement. We have  $\binom{4}{2} = 6$  pairs so

$$U_{\text{tot}} = 6U = -\frac{6Gm^2}{a}$$

The answer is D.